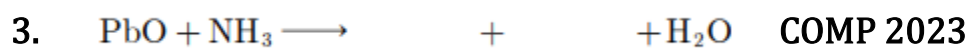
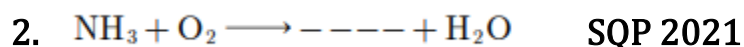


Chapter

Ammonia



5. Nitrogen gas can be obtained by heating : SQP 2023

- (a) Ammonium nitrate
- (b) Ammonium chloride
- (c) Magnesium nitride
- (d) Ammonium nitrite

6. During laboratory preparation how is ammonia dried and collected?
COMP 2021

7. State what do you observe when neutral litmus solution is added to an alkaline solution. SQP 2001

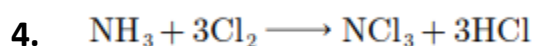
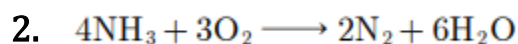
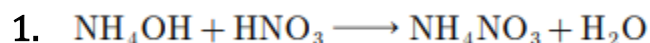
8. List the properties of ammonia that make it : COMP 2023

- (i) A good refrigerant
- (ii) A cleaning agent
- (iii) As a source of hydrogen.

9. (i) Ranjana wants to prove that ammonia is a reducing agent. To demonstrate this, she passes ammonia gas over heated copper oxide. What will she observe?

(ii) Write a balanced chemical equation for the above reaction. SQP 2025

Solution



5. (d) Ammonium nitrite

Nitrogen gas can be obtained by heating ammonium nitrite (NH_4NO_2). Upon heating, it decomposes to produce nitrogen and water:



6. In the laboratory preparation of ammonia, the gas is **dried by passing through quicklime (CaO)**, which absorbs moisture. It is then **collected by downward displacement of air** because ammonia is **lighter than air**.

7. It turns blue.

8. (i) **As a good refrigerant:** Ammonia can be **liquefied easily** and has a **high latent heat of vaporisation**, making it effective for **absorbing and removing heat**.

(ii) **As a cleaning agent:** Ammonia solution **emulsifies fats and grease**, which makes it useful for **cleaning purposes**.

(iii) **As a source of hydrogen:** Ammonia can be **liquefied and transported easily**, and on decomposition it **produces hydrogen** safely, whereas transporting **liquid hydrogen directly is dangerous**.

9. (i) **Observation:** When **ammonia gas** is passed over **heated copper(II) oxide (CuO)**, the **black CuO changes to reddish-brown/pink copper (Cu)**. This shows that **ammonia acts as a reducing agent**.

(ii) Balanced chemical equation:

